```
clear;
clc;
close all;

% Design Parameters
Fs = 10000;
Ap = 0.5;
As = 30;

% 1. Low-Pass Filter Design
fp_lp = 1500;
fs_lp = 2000;

wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);

[N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');

% 2. High-Pass Filter Design
fp_hp = 2500;
fs_hp = 1800;

wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);

[N_hp, Wn_hp] = cheb1ord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = cheby1(N_hp, Ap, Wn_hp, 'high');

% Display Numerical Outputs
fprintf('=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===\n\n');
```

```
=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===
```

```
fprintf('--- CHEBYSHEV LOW-PASS FILTER ---\n');
```

```
--- CHEBYSHEV LOW-PASS FILTER ---
```

```
fprintf('Filter Order (N_lp): %d\n', N_lp);
```

```
Filter Order (N_lp): 6
```

```
fprintf('Passband Edge Frequency: %.2f Hz\n', Wn_lp*(Fs/2));
```

```
Passband Edge Frequency: 1500.00 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_lp);
```

```
    0.0006    0.0038    0.0095    0.0127    0.0095    0.0038    0.0006
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_lp);
```

```
    1.0000   -3.7650    6.8198   -7.3274    4.8717   -1.8927    0.3368
```

```
fprintf('--- CHEBYSHEV HIGH-PASS FILTER ---\n');
```

```
--- CHEBYSHEV HIGH-PASS FILTER ---
```

```
fprintf('Filter Order (N_hp): %d\n', N_hp);
```

```
Filter Order (N_hp): 6
```

```
fprintf('Passband Edge Frequency: %.2f Hz\n', Wn_hp*(Fs/2));
```

```
Passband Edge Frequency: 2500.00 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_hp);
```

```
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_hp);
```

```
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650
```

```
% Frequency Response Calculation
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);

% Magnitude in dB
mag_lp = 20*log10(max(abs(H_lp), 1e-10));
mag_hp = 20*log10(max(abs(H_hp), 1e-10));

% Plotting
figure('Name', 'Chebyshev Type-I LPF and HPF Responses');

subplot(2,2,1);
plot(f, mag_lp, 'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev LPF Magnitude (Order N = ', num2str(N_lp), ')']);
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');
ylim([-60 5]);

subplot(2,2,2);
plot(f, angle(H_lp)*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev LPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

subplot(2,2,3);
plot(f, mag_hp, 'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev HPF Magnitude (Order N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);

subplot(2,2,4);
plot(f, angle(H_hp)*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev HPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

